



Multi-task Ant Colony Optimization for Multi-vehicle Path Planning

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Abstract

The Multi-vehicle Path Planning (MVPP) problem involves simultaneously solving multiple vehicle path planning tasks within a transportation network that may share similarities in space, tasks, or constraints. Existing methods for the MVPP problem mainly depend on single-task optimization approaches like the traditional ant colony optimization, which typically use either centralized pooling for collaborative optimization or sequential independent optimization of each vehicle's path. However, as the size of the transportation network expands, single-task optimization faces increasing challenges in solving the MVPP problem. Multi-task optimization is an emerging and promising field that aims to enhance the performance of optimization algorithms through multi-task learning and knowledge sharing, thereby positively impacting various applications like the MVPP. This paper proposes a Multi-Task Ant Colony Optimization (MTACO) method, which allows multiple ant colonies to explore their respective pheromone matrices, thereby facilitating implicit knowledge sharing. In MTACO, each ant colony is responsible for a single vehicle path planning task, and knowledge transfer between tasks is achieved through an archive-based initialization strategy and a probabilistic knowledge transfer strategy, which leverage similarities between tasks to improve overall optimization performance. Furthermore, a similar-task association strategy is introduced to assess the correlation between different vehicle path planning tasks, enabling more effective sharing of valuable search information. Experimental results demonstrate that MTACO outperforms several state-of-the-art methods in various test instances, especially when dealing with multiple vehicle path planning tasks with similarities, exhibiting significant advantages and delivering globally optimal or near-optimal path solutions for all tasks even in large-scale networks.

Keywords Multi-task optimization · Ant colony optimization · Vehicle path planning

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1 Introduction

Multi-task optimization (MTO) is a process of simultaneously optimizing multiple tasks within a single optimization framework [1]. Unlike traditional single-task optimization, MTO considers the trade-offs and synergies among multiple related tasks, aiming to find solutions that yield optimal or near-optimal results for each task. Though different tasks may have different optimization objectives, constraints, and properties, valuable knowledge can still be shared by leveraging their relationships, leading to enhanced overall efficiency space [2]. The idea of MTO has been introduced in the field of evolutionary computation, giving rise to a promising paradigm known as evolutionary multi-task optimization (EMTO) [3]. EMTO algorithms facilitate mutual promotion among tasks, thereby improving the optimization

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performance of evolutionary algorithms. Numerous EMTO algorithms have been developed in the last decade. The well-known Multi-Factorial Evolutionary Algorithm (MFEA) utilizes individuals as information bridges for knowledge exchange [3]. Experimental studies have demonstrated the superior search efficiency of MTO over traditional evolutionary algorithms.

The Multi-Vehicle Path Planning (MVPP) is a challenging combinatorial optimization problem that involves finding optimal routes for multiple vehicles. Each vehicle needs to determine the best path from origin to destination. Existing studies suggest that EMTO has shown promising results in solving combinatorial optimization problems, indicating its potential to solve MVPP problems [4, 5]. However, when it comes to MVPP, a challenge emerges in transferring knowledge between different tasks using traditional crossover operators. This difficulty arises because directly combining two candidate paths may result in infeasible paths for the vehicles since they often have different origin-destination positions. Consequently, developing a new EMTO method to address the MVPP problem effectively is highly desirable.

Ant colony optimization (ACO) is a powerful metaheuristic search algorithm primarily employed for solving combinatorial optimization problems [6, 7]. ACO has gained widespread recognition and been successfully applied to various fields, including vehicle path planning problems [8–10]. Although most existing ACO variants perform well on single-task vehicle path planning, they tend to have high computation overhead for generating satisfactory solutions when dealing with multiple vehicle path planning tasks simultaneously [11–14]. These studies often neglect the potential benefits of collaboration among ant colonies and the value of knowledge sharing between tasks.

To address the above issues, this paper proposes a Multi-Task Ant Colony Optimization (MTACO) for solving the MVPP problem. MTACO employs multiple ant colonies to tackle multiple tasks simultaneously by facilitating sharing pheromone matrices to construct feasible solutions. MTACO adopts an implicit migration pattern inherent in MVPP, whereby ACOs operate within the same search space (i.e., the same transportation network). The pheromone matrix of an ant colony can be dynamically selected based on task similarity using a probabilistic knowledge transfer strategy. The degree of similarity between tasks is assessed by a similar-task association strategy with a filtering technique. Similarity between different tasks is an important concept in the field of multi-task optimization. Similarity refers to the degree of association or interrelationship between different tasks, which can be used to describe the similarities or differences between tasks [2]. Tasks with greater similarity may bring more effective knowledge to other tasks. In addition, there is an archive storing valuable information from completed tasks, including pheromone matrixes and task features. The

pheromone matrix of an initial ant colony can be initialized as the pheromone matrices of similar tasks in the archive. Consequently, valuable knowledge can be derived not only from ant colonies performing similar tasks but also from ant colonies accomplishing similar tasks.

The main contributions of this paper can be summarized as follows.

1. We propose a multi-task ACO framework, named Multi-Task Ant Colony Optimization, with implicit knowledge transfer to address the MVPP problem.
2. Implementation of implicit knowledge transfer using pheromone matrices, leveraging information from ongoing and archived similar tasks.

The rest of the paper is organized as follows. Section 2 provides a review of related research on this topic. Section 3 presents the mathematical model for the MVPP problem. Section 4 offers a detailed description of the proposed MTACO approach. Experimental results and analysis are given in section 5 and section 6. Finally, the conclusion of this paper is drawn in section 7.

2 Related Work

2.1 Ant Colony Optimization Metaheuristic

ACO, like Ant Colony System (ACS), has been formalized as a metaheuristic for combinatorial optimization problems. A metaheuristic can be seen as a methodology that includes master strategies capable of guiding the search for the globally optimal solution [15]. ACO has four main steps: 1) randomly initializing ants and pheromones, 2) ants construct solutions based on pheromones, 3) update of local pheromones, and 4) update of global pheromones. The following is a more detailed description of the three phases.

2.1.1 Solution Construction

A set of m artificial ants select vertices from V to construct the solutions, where $V = \{v_i \mid i = 1, \dots, n\}$ is the set of vertices. Each ant is placed at the origin o_i and moves toward the currently reachable vertex $v_j \in V$ until the ant reaches the destination. The process of constructing a solution by an ant can be viewed as a single trip of a vehicle on a connected directed graph $G = (V, E)$, where $E = \{e_{ij} \mid i, j = 1, \dots, n\}$ is the set of edges representing connections between different vertices. The selection of the next vertex n from V is guided by a stochastic rule (e.g., Eq. (1)) that is influenced by the distances and the pheromone concentration of the path. Different ACO algorithms have

different rules for the random selection of vertices.

$$\omega = \begin{cases} \arg \max_{v_j \in \mathbf{J}_k(v_i)} \left\{ [\tau(v_i, v_j)] \cdot [\eta(v_i, v_j)]^\beta \right\}, & \text{if } q \leq q_0 \\ N, & \text{otherwise.} \end{cases} \quad (1)$$

Equation (1) is an example of a stochastic rule, called a state transition rule. An ant positioned at vertex v_i chooses the city ω to move to by applying the rule, where $\tau(v_i, v_j)$ is the pheromone concentration between v_i and v_j , $\eta(v_i, v_j)$ is the inverse of the distance ($1/w_{ij}$), $\mathbf{J}_k(v_i)$ is the set of vertices reachable by ants located on v_i , β indicates the importance of distance for ants, q is a random number uniformly distributed in $[0, 1]$, q_0 is a parameter ($0 \leq q_0 \leq 1$), and N is a random variable selected according to the probability distribution given in Eq. (2).

$$pr(v_i, v_j) = \begin{cases} \frac{[\tau(v_i, v_j)] \cdot [\eta(v_i, v_j)]^\beta}{\sum_{v \in \mathbf{J}_k(v_i)} [\tau(v_i, v)] \cdot [\eta(v_i, v)]^\beta}, & \text{if } v_j \in \mathbf{J}_k(v_i) \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

The $pr(v_i, v_j)$ is the probability that the vertex v_j will be selected for vertex v_i .

2.1.2 Update Local Pheromones

The update of local pheromones is to increase or decrease the concentration of pheromones on the paths walked by the ants. The purpose of increasing is to improve the probability that a good or promising path will be selected. The purpose of reducing is to allow the ants to better explore new paths (e.g., the local updating rule of ACS). Most of the ACO algorithms choose to increase the pheromone concentration to update the local pheromone.

2.1.3 Update Global Pheromones

The update of global pheromones simulates the volatilization of pheromones of natural ant colonies over time, the pheromones of all paths volatilize (decrease) in a certain proportion [16]. Used in conjunction with local updates, update strategy for global pheromones lowers the probability of poor paths being selected, allowing the ants to gradually converge on a good path.

2.2 Single-task and Multi-task Optimization

The primary difference between Single-Task Optimization (STO) and Multi-Task Optimization (MTO) lies in their approaches to handle multiple tasks. STO employs an independent optimization approach, where each task is

solved in isolation using only task-specific data without sharing information. For example, when solving multiple path planning problems, STO requires constructing a separate optimization model for each task and executing the optimization process independently. Though simple to implement for standalone tasks, this approach may lead to computational redundancy and low optimization efficiency for similar tasks. Taking an example of the regional logistics scheduling, STO requires building an independent scheduling plan for each region, ignoring potential overlaps in traffic patterns or delivery demands.

In contrast, MTO takes a smarter collaborative approach, where multiple related tasks are solved simultaneously within a unified optimization framework to leverage their similarities to improve the overall efficiency. MTO measures the similarity among multiple tasks and adopts either explicit or implicit knowledge transfer mechanisms, so that one task can use its optimization experience to help other tasks accelerate convergence and improve solution. For example, in a multi-robot path planning system, while individual robots have different destinations, they can share dynamic environmental knowledge such as obstacle locations and terrain constraints, thereby effectively pruning redundant state-space exploration. By pooling optimization experiences, MTO often finds better solutions with a faster convergence than STO, especially when tasks share common ground.

However, strength of MTO depends on task compatibility. If the differences between tasks are significant, knowledge transfer may result in negative transfer, meaning that the optimization experience from one task might adversely affect the convergence performance of another, thereby reducing overall optimization effectiveness [17].

2.3 Implicit and Explicit EMTO

Since EMTO has demonstrated notable performance, there have been many variants of EMTO in the past few years. Existing works for EMTO can be categorized into two types, i.e., implicit transfer models and explicit transfer models [2].

In works of the implicit type, knowledge transfer is implicitly achieved through evolutionary operators for individuals with different tasks, which allows individuals to perform crossover operations in the same search space. In this type, MFEA [3] is the most representative, which uses a single population in a unified search space, where each individual selects a task to cross-operate based on a skill factor, implicitly enabling knowledge transfer. Later, Bali et al [18] studied online transfer parameter estimation and proposed MFEA-II. Ding et al. [19] proposed a generalized MFEA with decision variable leveling and shuffling strategies. Li et al [1] proposed a multi-task evolutionary framework based on adaptive solver selection called MTEA-SaO, which is capable of assigning

multiple sub-populations to each task and selecting the most appropriate sub-population. Li et al. [20] proposed a differential evolutionary algorithm based on meta-knowledge transfer to solve the MTO problem more efficiently through a more generalized meta-knowledge transfer approach. Chen et al. [21] proposed a novel evolutionary framework, the many-task evolutionary algorithm, for many-task optimization. Xu et al. [22] proposed an adaptive EMTO framework to reduce the impact of negative transfer. The framework can synergistically adjust knowledge transfer frequency, knowledge source selection, and knowledge transfer intensity. Qiao et al. [23] proposed a double-balanced evolutionary multi-task optimization algorithm, which evolves two populations to respectively solve constrained multi-objective optimization problems and multi-objective optimization problems. Wu et al. [24] developed an effective PCR approach for the low success ratio problem of evolving registration. Wu et al. [25] modeled the registration problem as multi-task optimization, and proposed a novel bi-channel knowledge sharing mechanism for effective and efficient problem solving.

These works share a common goal of enhancing knowledge transfer in MTO frameworks to improve optimization efficiency. They focus on leveraging implicit knowledge transfer through evolutionary operators, as seen in MFEA and its extensions, where individuals working on different tasks naturally exchange information in a unified search space. Several approaches introduce adaptive strategies to fine-tune the knowledge transfer process, dynamically adjusting factors like transfer frequency and intensity to avoid negative transfer. Additionally, some methods utilize multi-population or sub-population structures to assign specific populations to tasks, ensuring better-suited interactions between tasks. Lastly, meta-knowledge transfer and generalized frameworks provide flexible and robust ways to transfer knowledge across diverse tasks, enabling more efficient problem-solving across complex optimization scenarios.

On the other hand, the second type of MTO works focuses on explicit knowledge transfer between different populations. Different populations have different search spaces. Knowledge transfer between populations occurs at the end of each iteration, which uses feasible solutions to transfer. For example, Feng et al. [4] proposed an explicit genetic transfer method based on automatic coding for knowledge transfer between populations targeting different tasks. Du et al. [26] proposed a more aggressive transfer by transferring valuable solutions and proposed an algorithm for a multi-objective multi-task optimization algorithm. Wu et al. [27] proposed an orthogonal transfer method to deal with heterogeneous search space and dimensional similarity to achieve high-quality transfer. Jiang et al. [28] proposed block-level knowledge transfer to enhance knowledge transfer between

the same or different task dimensions. Moreover, in the field of swarm intelligence algorithm, Chen et al. [29] proposed a particle swarm algorithm-based feature under-selection method to solve the high-dimensional classification problem by sharing information between two related tasks. Tang et al. [30] devised a novel level-based inter-task learning strategy upon a dynamic local topology of inter-task particles, which separates particles into several levels and assigns particles to different levels with distinct inter-task learning methods. These research works show that MTO can enhance the performances of both evolutionary and swarm intelligence algorithms.

These works share a common focus on explicit knowledge transfer between populations in MTO. Unlike implicit methods, they clearly define how knowledge is exchanged between populations operating in different search spaces, with transfer typically occurring at the end of each iteration using high-quality or feasible solutions. Several methods address the challenge of heterogeneous search spaces by ensuring that knowledge transfer between populations is effective, even when task dimensions differ, such as through orthogonal transfer or block-level knowledge sharing. Additionally, these explicit transfer strategies are applied to both evolutionary and swarm intelligence algorithms, with examples like particle swarm optimization demonstrating the broad applicability of these approaches. Overall, these methods enhance optimization performance by directly exchanging valuable solutions between populations, thereby improving the efficiency and effectiveness of multi-task learning.

The MTACO proposed in this paper also belongs to the type of implicit knowledge transfer, where multiple vehicle path planning tasks are in the same transportation network. Thus, multiple ant colonies can generate candidate paths in the unified search space and they can have pheromone matrices of the same size, which provides the basis for knowledge transfer.

2.4 ACO for Vehicle Path Planning

As a classical algorithm for combinatorial optimization, ACO has been used in many works to solve vehicle path planning problems [31–33]. Jabir et al. [34] proposed an efficient algorithm based on ACO and Variable Neighborhood Search to solve multi-depot green vehicle routing problems with the objective of minimizing cost and emission. Wang et al. [35] proposed a novel hybrid ACO-based algorithm in which ants can enter and exit the warehouse in multiple rounds. The model was then applied to the combination of cumulative vehicle routing problems and multi-depot vehicle routing problems. Manogaran et al. [13] proposed a route optimization method based on a fitness-ant colony optimization to improve the electric vehicle range by controlling delays and

eliminating electric vehicle failures caused by power exhaustion. Xiang et al. [12] proposed a meta-heuristic algorithm combining a demand coverage diversity adaptive approach with ACO, called ACO-CD, for solving dynamic vehicle path planning problems. Despite the extensive research on ACOs for vehicle path planning, the work of ACOs to solve the MVPP problem is very rare.

These works share a common focus on utilizing ACO to tackle various VPP problems, often integrating additional techniques to enhance its performance. The widespread application of ACO in VPP, but its use in MVPP is still underexplored. MTACO has significant potential for MVPP, as it could leverage the strengths of ACO while facilitating knowledge sharing across multiple tasks. This approach could lead to more efficient and scalable solutions for complex transportation networks, allowing for the simultaneous optimization of multiple vehicle routes and improving overall traffic management.

2.5 ACO for Multi-task Optimization

Compared to the active research of EMTO, ACO has rarely been studied for multi-task optimization. With the help of heuristic information, ACO may have better results than evolutionary algorithms for many combinatorial optimization problems. Therefore, it is promising to introduce the idea of multi-task optimization into ACO. Recently, a novel multitasking ant colony optimization (named MTACO-DMSI) to detect multi-order single-nucleotide polymorphism (SNP) interactions have been proposed by Tuo et al [36]. The MTACO-DMSI algorithm performs multiple multi-order SNP interaction detection tasks in parallel during the search phase and generates two separate sub-populations for each task. Since its tasks carry fixed correlations with each other, MTACO-DMSI does not need to evaluate the similarity between tasks. MTACO-DMSI uses explicit knowledge transfer based on elite solutions, i.e., the optimal solution constructed by an ant colony is transformed into feasible solutions for other ant colonies.

MTACO-DMSI demonstrates that explicit knowledge transfer between tasks, based on elite solutions, can be effective for problems where task similarities are fixed, as in multi-order SNP interaction detection. In such cases, tasks are closely related, allowing direct solution sharing between ant colonies.

However, the MVPP introduces unique challenges, such as uncertain and dynamic task similarities. Unlike SNP detection, where task relationships are fixed, vehicle routing tasks can vary significantly in scale, constraints, and objectives. This makes direct application of MTACO-DMSI's knowledge transfer strategy difficult. Nevertheless, the prospects of MTACO for MVPP remain promising. By developing adaptive knowledge transfer mechanisms that account for

dynamic task similarities and addressing the challenges posed by diverse solution spaces, MTACO can potentially offer more efficient and scalable solutions for complex, large-scale transportation networks.

Specifically, MTACO is preferred for the MVPP problem due to three following advantages of its implicit knowledge transfer mechanism. The first advantage is reduced structural conflicts. Explicit knowledge transfer operates at the solution space level, requiring consistent encoding methods across tasks, which can lead to infeasible solutions when solutions are combined or exchanged. By contrast, implicit transfer shares environmental experience (i.e., pheromones) rather than specific solution structures. This prevents direct conflicts between different task solutions and ensures that generated paths remain feasible and valid for each task. The second advantage is preservation of solution diversity. Explicit transfer usually involves directly merging individual solutions, which may cause premature convergence. On the contrary, implicit knowledge transfer modifies pheromone concentration dynamically, which indirectly affects ants' path selection. This mechanism allows ants to retain their original search characteristics while gradually converging towards the best solutions discovered in other tasks, thereby maintaining population diversity. The last advantage is reduced risk of negative Transfer. Explicit knowledge transfer typically involves directly copying or crossbreeding elite solutions, which requires the solution structures of different tasks to be completely consistent. In multi-vehicle path planning, tasks may be somewhat similar, but variations in starting points, destinations, and specific route structures can lead to incompatibility. Directly transferring solutions may introduce inappropriate paths, leading to negative transfer that reduces overall search efficiency. In contrast, implicit knowledge transfer via soft pheromone sharing only influences ants' decision-making probabilistically without forcibly modifying their search behaviors. This approach preserves the individuality of different tasks while still allowing successful paths to be leveraged across tasks.

3 Problem Formulation

The vehicle path planning problem (VPP) is a real-world manifestation of the shortest path problem, typically aimed at providing the driver with the optimal path from the origin to the destination, minimizing travel time, path length, or other objectives [37]. However, in the Multi-Vehicle Path Planning (MVPP) problem, the challenge is not only to find the optimal path for a single vehicle but to simultaneously plan the optimal paths for multiple vehicles. This significantly increases the complexity of the problem, especially in real transportation networks where MVPP must consider more constraints, such as traffic conditions, road types, speed limits, and more.

These constraints make it even more difficult for traditional methods to find optimal solutions in a reasonable time, often classifying MVPP as an NP-hard problem.

Similar to VPP, MVPP can also be modeled as a connected directed graph $G = (V, E)$, where $V = \{v_i \mid i = 1, \dots, n\}$ is the set of vertices and $E = \{e_{ij} \mid i, j = 1, \dots, n\}$ is the set of edges representing connections between different vertices. Each edge e_{ij} is assigned a non-negative weight w_{ij} , which may represent energy consumption, link length, time cost, or a combination of these. In this paper, the weight w_{ij} is defined as the length of the links. In MVPP, multiple vehicles start from different source vertices and aim to reach their respective destination vertices. The goal of the algorithm is to find the optimal paths for each vehicle simultaneously while minimizing the total weight of all the vehicle paths.

Therefore, MVPP can be viewed as an extended and more complex version of VPP, where handling the path planning tasks of multiple vehicles introduces additional dimensions and challenges. This paper evaluates the proposed algorithm through unconstrained MVPP to demonstrate its effectiveness in simultaneously planning paths for multiple vehicles.

$$\min f(x) = \sum_{i,j \in \{1, \dots, n\}, i \neq j} W(x_{ij}), \quad (3)$$

Equation (3) shows the objective of the VPP to minimize the total traveling distance of the vehicle, where $f(x)$ is the fitness function that is the total distance traveled and $x = [v_o, \dots, v_e]$ represents a feasible solution to the task t , where v_o denotes the origin vertex of the path and v_e denotes the end vertex of the path. $W(x_{ij})$ denotes the distance from the i -th vertex to the j -th vertex recorded in the feasible solution x .

The multiple VPPs have K tasks to be solved and each task has the same form of fitness function. Then MVPPs aim to find a set of feasible solutions $\{x_1, x_2, \dots, x_K\}$, which can be systematically stated as follows.

$$(x_1^*, x_2^*, \dots, x_K^*) = \arg \min(f_1(x_1), f_2(x_2), \dots, f_K(x_K)) \quad (4)$$

where x^* is the optimal feasible solution of the task and K is the total number of tasks.

4 Multi-task Ant Colony Optimization

The framework of MTACO is shown in Fig. 1. MTACO utilizes multiple ant colonies to solve multiple tasks, with a single ant colony for a specific task. Each task is handed over to a single colony ACO. In MTACO, ant colonies with similar tasks can exchange their experiences, which are not limited to ongoing ant colonies or ant colonies that have already finished their iterations. Single colony ACOs play the role

of task solvers in MTACO. To better explain the knowledge transfer strategy of MTACO, several terms are defined. The ant colony that receives the knowledge is called the target ant colony. The one that provides the knowledge is referred to as the source ant colony. Ant colonies with similar tasks are called similar ant colonies. Ant colonies with tasks already accomplished are known as archived ant colonies. Each ant colony records a pheromone matrix and task features (origin and destination).

First, a similar task association strategy is applied to identify relationships between similar ant colonies. Ant colonies a_j , which are similar to the target ant colony a_t , are recorded in the set A_t^s . This set contains all similar ant colonies related to the target ant colony a_t . With several archived ant colonies stored in an archive beforehand, an archive-based initialization strategy was used to obtain useful information from archived ant colonies with the most similar tasks. The archive-based initialization strategy initializes the pheromone matrix of the target ant colony based on the extracted information to explore more promising paths when the task begins. For similar ant colonies, a probabilistic knowledge transfer strategy is proposed to select a pheromone matrix from source ant colonies for a target ant colony before calculating the state transfer probabilities whenever in need. The idea behind the above strategies is that the experience of ant exploration can be conveniently shared between ant colonies with the help of pheromone matrices.

4.1 Proposed Strategies

4.1.1 Similar-Task Association Strategy

Invalid knowledge will lead to inefficiencies in EMTO when tasks are very different, which is called negative migration. In order to avoid the impact of negative migration, a similar-task association strategy is specially designed for the MVPP problem in this paper, which contains two steps, namely the similar-task filtering and the similarity measurement.

Similar-task filtering is to find out the highly similar tasks to the target task. For MVPP, the correlation between tasks is judged by positions of the origins and the destinations. Fig. 2 illustrate the proposed strategy. There are four tasks t_1, t_2, t_3 , and t_4 to be solved, where task t_2 is the target task. The symbol o denotes the origin and e denotes the destination. The origin o_2 and destination e_2 is used as centers. If $d(o_2, o_i) \leq r$, task t_i is made as a candidate similar-task for the target task t_2 , where $d(o_2, o_i)$ denotes the distance from vertex o_i to o_2 , $i = \{1, 3, 4\}$, r is radius of the circle. Subsequently, task t_4 has been excluded and tasks t_1 and t_3 are considered candidates for task t_2 . Then, if $d(e_2, e_i) \leq r$, task t_i ($i = 1, 3$) is a similar task for the target task t_2 , where task t_i is a candidate similar-task. Eventually, task t_1 is a similar task to task t_2 .

Fig. 1 Framework of the MTACO

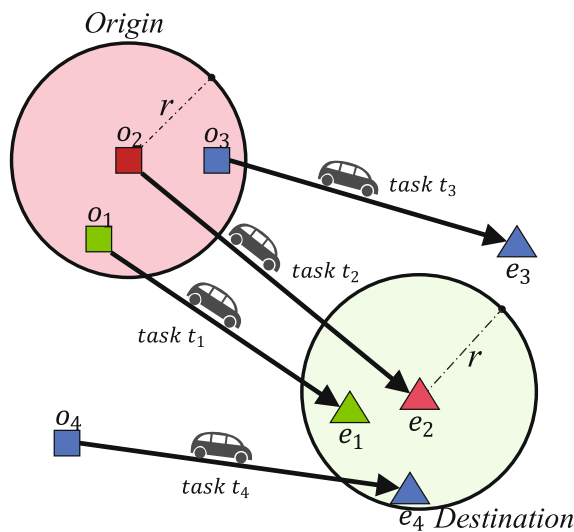
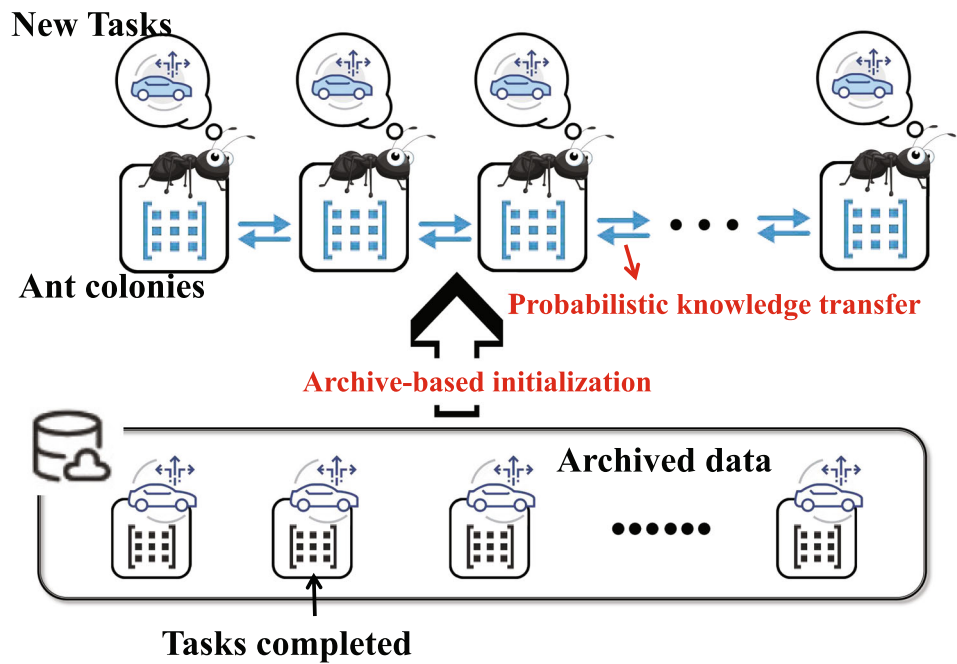


Fig. 2 The similar-task filtering for MVPP Problem

After judging whether the tasks are relevant, the similarity between the tasks needs to be calculated. The higher the similarity between tasks, the knowledge from the source ant colony more likely help the target ant colony to achieve better results.

The probability of task knowledge being helpful increases with similarities between tasks. The similarity can be calculated by

$$s^m(t_i, t_j) = \frac{1}{d(o_i, o_j) + d(e_i, e_j)}, i, j = 1, 2, \dots, K, i \neq j, \quad (5)$$

where $s^m(t_i, t_j)$ denotes the similarity value between task t_i and task t_j . The origin and destination positions between tasks are treated as elements of the similarity measure. The core idea of the strategy is that the close origins and destinations imply some overlaps of the paths of two vehicles. This means that they may share a similar navigational section.

Then, the ant colony a_j with task t_j is recorded in the set A_i^s of the target ant colony a_i , which is denoted as

$$A_i^s = \{a_i^j(p_j, s_{ij}^m) | j = 1, \dots, K, t \neq j, j \leq K, \quad (6)$$

where $s_{ij}^m = s^m(t_i, t_j)$, K is the number of tasks, $a_i^j(p_j, s_{ij}^m)$ records pheromone matrix p_j and its similarity s_{ij}^m to an ant colony a_i , a_i^j is an ant colony a_j that is similar to ant colony a_i , A_i^s records all similar ant colonies to an ant colony a_i .

4.1.2 Archive-based Initialization Strategy

In ACO, the positive feedback mechanism of pheromones simulates the behavior of ants releasing pheromones along their paths. Ants tend to choose paths with higher pheromone concentrations, as these paths have typically been visited by more ants and are likely to be better choices. To leverage historical optimization experience, an archive-based initialization strategy can be adopted, where different paths are assigned with varying levels of pheromone concentration before the search begins, guiding the ant colony toward high-quality solutions. In the archive-based initialization strategy, historical optimal paths are assigned with higher pheromone concentrations, while inefficient paths receive lower pheromone levels. This helps reduce early-stage ran-

dom searches and improves the quality of the initial solutions. However, not all historical solutions are beneficial for the current optimization task. If the archived ant colony solutions have low relevance to the target task, directly using these pheromones for initialization may introduce irrelevant information, potentially reducing search efficiency. Therefore, before initializing ACO, it is essential to evaluate the similarity between the archived ant colony and the current task and select the most relevant archive data for initialization.

To ensure that the archived pheromone values do not excessively influence the current task, the pheromone values from the archive are typically normalized to maintain the same magnitude as the standard initial pheromone concentration. Finally, the two pheromone matrices (the archive-based pheromone matrix and the baseline initialization pheromone matrix) are combined using weighted fusion. The weight of the two pheromone matrices is set at a 1 : 1 ratio, meaning that the archive-based pheromone matrix and the baseline initialization pheromone matrix contribute equally to the final pheromone distribution. This ensures a balance between leveraging historical high-quality solutions and maintaining exploration capability in the early stages of optimization. By equally incorporating prior knowledge and standard initialization, the algorithm can achieve faster convergence while avoiding excessive bias toward previously explored solutions. The initialization formula is given by

$$p_{ij}^t = p_{ij}^t + \frac{p_{ij}^h - p_{min}^h}{p_{max}^h - p_{min}^h} \times \varepsilon, i, j = 1, \dots, n, p_{max}^h \neq p_{min}^h, \quad (7)$$

where the variable p^h is the pheromone matrix of the archived ant colony selected and p^t is the pheromone matrix of the target ant colony. The variables p_{max}^h and p_{min}^h are the maximum and minimum values in the pheromone matrix p^h , respectively. Each element of the pheromone matrix p^h is linearly scaled to a range of values with close initial concentrations. This prevents a priori knowledge having too much influence on the target ant colony. The variable p_{ij} represents the element in row i and column j of the pheromone matrix. The ε is a constant to adjust the range of normalized data, whose value needs to be defined specifically based on the nature of the data and the need for initialization.

Note that the value setting of ε requires comprehensive consideration of the initial pheromone concentration, normalization effects on colony search behavior, and optimization process stability. If ε is excessively large, it may drastically exceed the initial pheromone level, causing drastic adjustments to the pheromone concentration that disrupt the colony's natural search patterns. This perturbation could lead to premature convergence by overriding the algorithm's inherent exploration mechanisms. Conversely, an undersized

ε value produces negligible updates during pheromone normalization, weakening the directional guidance for colony search and diminishing optimization efficiency. Proper calibration of ε therefore can help to prevent local optimum entrapment through controlled pheromone modulation, as well as to enable progressive pheromone accumulation that maintains an effective balance between exploration and exploitation within the search space.

4.1.3 Probabilistic Knowledge Transfer Strategy

As shown in Fig. 3, this strategy is responsible to transfer knowledge from the source ant colony to the target ant colony. The strategy uses a decision factor q_r to decide whether the ant colony uses its own pheromone matrix or the one from source ant colony, which can be expressed by

$$p = \begin{cases} p_s, & \text{if } rand(0, 1) \leq q_r, \\ p_o, & \text{otherwise} \end{cases} \quad (8)$$

where p is the pheromone matrix used to calculate the transfer probability. p_s is the pheromone matrix of the target ant colony. p_o is the pheromone matrix of the source ant colony, which is selected according to a pheromone pseudo-random selection strategy. The $rand(0, 1)$ denotes a random number from 0 to 1. q_r is the transfer decision factor that regulates the tendency of the target ant colony to choose the pheromone matrix. If $rand(0, 1) > q_r$, the knowledge transfer is performed. Otherwise, the target ant colony uses the pheromone matrix p_o to compute the state transfer probabilities.

The pheromone matrix p_o is given by the pheromone pseudo-random selection strategy, which comprises greedy selection and random selection methods, as expressed by

$$p_o = \begin{cases} \text{pheromone matrix of } \arg \max_{a_t^j \in A_t^s, t \neq j} S(a_t^j), & \text{if } rand(0, 1) \leq q_s \\ \text{roulette-wheel selection,} & \text{otherwise,} \end{cases} \quad (9)$$

where q_s determines the relative importance of greedy selection versus random selection, the $S(a_t^j)$ denotes task similarity s_{ij}^m between the target ant colony a_t to the similar ant colony a_j . Whenever the pheromone selection is required, it samples a random number $rand(0, 1)$. If $rand(0, 1) \leq q_s$, the target ant colony a_t will select the pheromone matrix of the source ant colony with the highest task similarity. The probability of the pheromone matrix being selected can be calculated by

$$P_i = \frac{s^m(t_i, t_i)}{\sum_{j=1}^k s^m(t_i, t_j)}, i, j = 1, \dots, k, \quad (10)$$

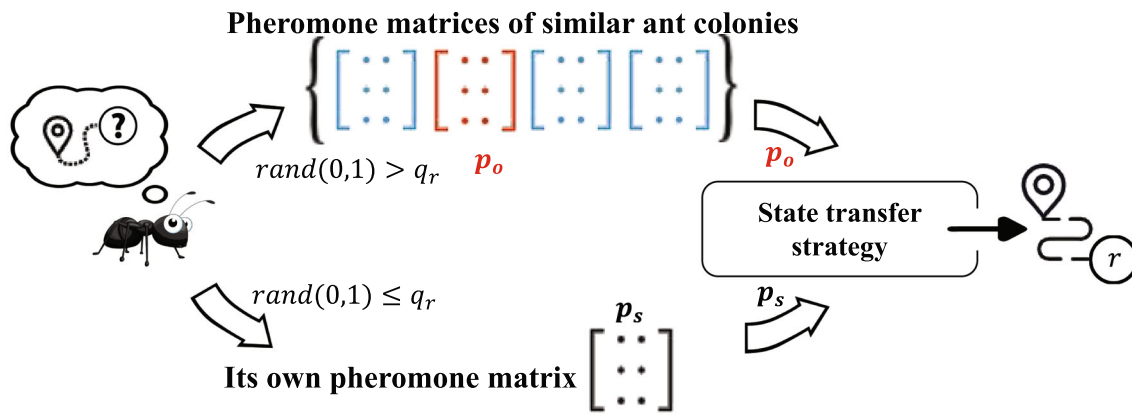


Fig. 3 Ants select vertices by probabilistic knowledge transfer strategy

where P_i is the probability that the pheromone matrix p_i of the source ant colony a_i being chosen, t_i is target task, and t_i or t_j is a similar task. Roulette-wheel selection randomly selects the pheromone matrix according to the calculated probability P_i .

4.2 Implementation Procedures

In this section, the proposed MTACO, with multiple single colony ACS, is specifically introduced to solve MVPP problems. The pseudo-code of MTACO is provided in Algorithm 1, which mainly contains three phases of the initialization, the iterative search, and the pheromone updating. Algorithm 2 describes the process of the ant exploring path.

4.2.1 Initialization

The goal of this phase is to prepare an initial ant colony before the iterative search process (Lines 1-20 of Algorithm 1). The quality and rationality of the initialization are very important for the performance of the algorithm. Assuming there are K VPP tasks, there will be K ACOs, which have their ant colony, pheromone matrix, and parameter settings, where the scale of the pheromone matrix is $|V| \times |V|$, with $|V|$ being the number of vertices on the transportation network, each ant colony performing a specific task. Then, these ant colonies obtain their respective similar ant colony set A^s according to the similar-task association strategy (Lines 3-13 of Algorithm 1). In other words, ant colonies with similar tasks will be stored in the same set. Then, the pheromone matrix of each ant colony can be initialized with the aid of archived ant colonies with similar tasks (Lines 14-20 of Algorithm 1).

Algorithm 1 MTACO Pseudo-code

Input: initial pheromone level τ_0 ; edge set E ; vertex set V ; task set T ; number of tasks K ; ants m ; radius r ; heuristic function weight β ; transfer probability decision factor q_r ; pheromone matrix selection weight q_s ; state transfer decision factor q_0 ; local pheromone evaporation rate ρ ; global pheromone evaporation rate α ; maximum iterations G ; ant colony set A ; archived set A^h .

Output: a set of feasible route X .

```

1: /* Initialization */
2: /*  $i, j \in [1, K]$  denote the index of the ant with task */
3: for  $i \in [1, K]$  do
4:   /* Filtering similar tasks */
5:   for  $j \in [1, K]$  do
6:     if  $i \neq j$  and  $o_i \neq o_j$  and  $e_i \neq e_j$  then
7:       if  $d(o_i, o_j) \leq r$  and  $d(e_i, e_j) \leq r$  then
8:          $A_i^s[j] = a_j^s(p_j, s_{ij}^m)$  /*  $A_i^s$  is an array of size  $K$  to record
           similar ant colonies */
9:       end if
10:    end if
11:  end for
12:  /* Select the source ant colony with the highest similarity to the
    target ant colony from the archived */
13:   $a_h = \arg \max S(a_j^h), a_j^h \in A^h$  /*  $a_i$  may not have the similar ant
    colony  $a^h$  in the archived set  $A^h$  */
14:  /* Initializing the pheromone matrix of the target ant colony */
15:  if  $\exists a_h$  then
16:     $p^h$  = the pheromone matrix of ant colony  $a_h$ 
17:     $p_i = \text{Archive-based Initialization Strategy}(p^h)$  (Eq. (7))
18:  end if
19:  Append  $a_i(t_i(o_i, e_i), \tau_0, m, \beta, q_0, \rho, \alpha, p_i, A_i^s)$  to  $A$ 
20: end for
21: /* Iterative search. */
22:  $g = 1$ 
23: while  $g \leq G$  do
24:   for  $a_i \in A$  do
25:     for  $c \in [1, m]$  do
26:       /*  $ant_c$  constructs a feasible solution  $x_c$  */
27:       The  $ant_c$  begins at origin  $o_i$  and explores the path (Algo-
       rithm 2)
28:       Local Updating Rule( $p_i, \rho, x_c$ )
29:     end for
30:     Global Updating Rule( $p_i, \alpha, x^{best}$ )
31:   end for
32: end while
33: return  $X = \{x_1, \dots, x_K\}$ 

```

Algorithm 2 Path construction for single task

Input: edge set E ; vertex set V ; number of ants m ; heuristic function weight β ; transfer probability weight q_r ; pheromone matrix selection weight q_s ; state transfer decision factors q_0 ; the similar ant colonies set A_i^s .

Output: a feasible route x .

```

1: while ant is not at destination  $e_i$  do
2:   /* Probabilistic knowledge transfer */
3:   if  $\text{rand}(0, 1) \leq q_r$  then
4:      $p = p_s$  /* Own pheromone matrix */
5:   else
6:     if  $\text{rand}(0, 1) \leq q_s$  then
7:        $p_o = \arg \max S(a_i^j), a_i^j \in A_i^s$  /* The pheromone matrix of
         the source ant colony */
8:     else
9:       The  $p_o$  obtained based on roulette-wheel selection
10:    end if
11:     $p = p_o$ 
12:  end if
13:  /* State transition */
14:   $v = \text{State Transition Strategy}(\beta, V, E, q_0, p)$ 
15:  ant go to  $v$  from current position
16: end while
17: return  $x = \{o_i, \dots, e_i\}$  ( $x$  is a set of vertices representing a complete path)

```

4.2.2 Iterative Search

Each ant repeats the following steps to construct a path (i.e., a feasible solution to the VPP task). Fig. 4 takes the task $t(v1, v4)$ from origin vertex $v1$ to destination vertex $v4$ as an example. In the third iteration, the ant is located at vertex $v5$ and the path is $\langle v1, v3, v5 \rangle$. The vertices connected to vertex $v5$ are $\{v2, v6, v7\}$. Based on the pheromone matrix obtained from the probabilistic knowledge transfer strategy, the ant can select vertexes based on a pseudo-randomized state transfer strategy. Assuming that vertex $v6$ is selected, the path constructed is updated to $\langle v1, v3, v5, v6 \rangle$. The ant repeats the above steps until it reaches the vertex $v4$. Then, after the ant constructs a path, the pheromone concentrations of the visited edges are modified via the local updating rule. All steps of the iterative search phase are described in detail below.

Step 1 Knowledge transfer: This step is performed before the ant travels to the next vertex. The ant colony selects either the pheromone matrix p_o of source ant colonies or its own pheromone matrix p_s for the state transfer probability calculation. The selection depends on the transfer decision factor q_r , whose value determines how much the target ant colony depends on its own pheromone matrix p_s . Usually, it is recommended for the target ant colony to refer to other pheromone matrices p_s with a small probability. In MTACO, the roulette random selection strategy is used to select other pheromone matrices based on task similarity, according to Eq. (10).

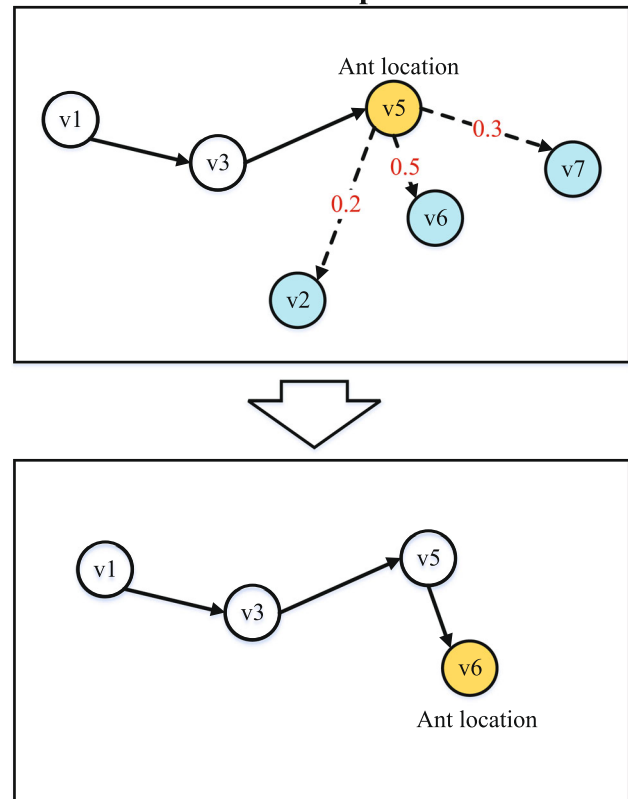
State transfer process

Fig. 4 An example of the state transfer in an ant, with state transfer probabilities marked in red

Step 2 State Transition Strategy: The probability of an ant traveling from the positioned vertex r to the next reachable vertex s can be calculated by

$$p_k(r, s) = \begin{cases} \frac{\tau(r, s) \cdot [\eta(r, s)]^\beta}{\sum_{u \in J_k(r)} \tau(r, u) \cdot [\eta(r, u)]^\beta}, & \text{if } s \in J_k(r) \\ 0, & \text{otherwise,} \end{cases} \quad (11)$$

$$\eta(r, s) = \frac{1}{w_{rs}} \quad (12)$$

where $\eta(r, s)$ is the heuristic information, setting as the reciprocal of the link length w_{rs} from vertex r to vertex s . Meanwhile, $\tau(r, s)$ is the pheromone concentration, and $J_k(r)$ is the set of vertices linked to vertex r . The probability of a reachable vertex being selected is calculated according to the transfer probability in Eq. (11). Then the next vertex to travel is selected according to the pseudo-randomized proportionality rule, as expressed by

$$s = \begin{cases} \arg \max_{u \in J_k(r)} p_k(r, u), & \text{if } \text{rand}(0, 1) \leq q_0 \\ \text{roulette-wheel selection,} & \text{otherwise.} \end{cases} \quad (13)$$

If a randomly generated number within $[0, 1]$ is less than or equal to the parameter q_0 , the vertex with the highest probability is chosen. Otherwise, a vertex is randomly selected. The probability for roulette-wheel selection is given by Eq. (11).

In the MVPP problem, though the heuristic information is fixed, the pheromone concentration of different pheromone matrices on the same path may be different. Therefore, the pheromone matrix plays a key role to guide ants to explore paths in the search space. Higher pheromone concentrations usually indicate better paths, which attract ants more to choose. This guiding helps the ants to focus on paths that have performed well in previous iterations, thus increasing the chances of finding a good solution. Overall, the pheromone matrix acts as a medium of shared experience in the ant colony, reflecting the results of previous iterations and guiding future searches through the behaviors of ants. In MTACO, the scope of sharing is extended to different ant colonies.

4.2.3 Pheromone Updating

The pheromone updating rules in ACS include the local updating rule and global updating rule respectively. The local updating rule allows all the ants release pheromones after exploring feasible paths, while the global updating rule only allows the best ants (i.e., those constructing the shortest paths so far) to release pheromones.

Step 1 ACS Local Updating Rule: After acquiring a solution (i.e., a feasible path from the origin to the destination) to the task, ants change pheromone levels at edges traveled by applying the local updating rule, as expressed by

$$\tau(r, s) = (1 - \rho) \cdot \tau(r, s) + \rho \cdot \tau_0, \quad (14)$$

where $0 < \rho < 1$ is a pheromone volatilization rate, τ_0 is initial level of pheromones.

Step 2 ACS Global Updating Rule: The global updating rule is executed after all ants have completed their journey. The pheromone concentration is adjusted by applying the global updating rule:

$$\tau(r, s) = (1 - \alpha) \cdot \tau(r, s) + \alpha \cdot \Delta\tau(r, s), \quad (15)$$

$$\Delta\tau(r, s) = \begin{cases} (L_{gb})^{-1}, & \text{if } (r, s) \in x^{best} \\ 0, & \text{otherwise,} \end{cases} \quad (16)$$

where the $0 < \alpha < 1$ is a pheromone volatilization rate, $(L_{gb})^{-1}$ is the historically best path. The global updating rule only updates the pheromone concentration of edges (r, s) belonging to the best path x^{best} .

4.3 Time Complexity Analysis

Time complexity analysis is essential for evaluating the computational efficiency of the MTACO algorithm. Given an MVPP with multiple tasks, each task requires to find an optimal path within the same transportation network. The complexity of MTACO primarily depends on the number of ants, the number of tasks, the network size, and the number of iterations. To provide a structured comparison, we first analyze the computational complexity of ACO, then derive that of MTACO, and finally compare the two approaches.

In a standard ACO algorithm, each ant constructs a solution by iteratively selecting vertices based on pheromone values and heuristic information. In the worst case, each vertex has multiple neighbors, leading to a computational cost of $O(|V|^2)$ per ant. Given that there are N ants per iteration, the total complexity for path construction is $O(N|V|^2)$ per iteration.

The pheromone update process consists of local updates by each ant and global updates applied at the end of each iteration. Local updates require modifying edge weights in each ant's route, leading to a cost of $O(N|V|)$. Global updates involve modifying the best solution's pheromone matrix, adding a cost of $O(|E|)$, resulting in a total pheromone update complexity of $O(N|V| + |E|)$. Given that ACO runs for G iterations, the total time complexity per run is $O(G(N|V|^2 + N|V| + |E|))$. Since $|E|$ and $N|V|$ are typically dominated by $N|V|^2$, this simplifies to $O(GN|V|^2)$.

MTACO extends ACO by incorporating multi-task optimization mechanisms, which introduce additional computational overhead. The key differences arise from solution construction, pheromone updating, archive initialization, task similarity evaluation, and probabilistic knowledge transfer.

During the solution construction phase, each ant follows the same vertex selection process as ACO. However, since MTACO handles M tasks simultaneously, with N ants per task, the total complexity for path construction in one iteration becomes $O(NM|V|^2)$.

The pheromone update phase involves local and global updates. In the path construction phase, each ant usually constructs a complete path that needs to pass through all (or most) of the vertices in the network, so its number of edges is approximately $|V| - 1$, approximated as $|V|$. Local updates require modifying edge weights in each ant's route, leading to $O(NM|V|)$. The global update step involves modifying the best solutions across all tasks, contributing an additional cost of $O(M|E|)$. This results in a total pheromone update complexity of $O(M|E| + NM|V|)$.

MTACO introduces an archive initialization phase. Each task compares to M_A stored tasks to determine similarity, with the complexity of $O(MM_A)$. Task similarity evaluation is another additional step in MTACO, requiring pairwise

Table 1 Test instance data, including mean similarity between tasks, standard deviation of similarity, and transportation network

Instance	Tasks	Similarity mean	std.	Similar Tasks	Network	Vertices	Links
T_1	10	0.0469	0.0138	9	$Gz-50$	50	146
T_2	30	0.0262	0.0065	27			
T_3	50	0.0209	0.0040	49			
T_4	10	0.0364	0.0043	9	$M-200$	200	874
T_5	30	0.0261	0.0032	29			
T_6	50	0.0228	0.0027	49			

comparisons among M tasks, resulting in a complexity of $O(M^2)$. Sorting and filtering the most relevant tasks add an overhead of $O(M \lg M)$, bringing the total similarity evaluation complexity to $O(MM_A + M^2 + M \lg M)$. Assuming $M_A \leq M$, both MM_A and $M \lg M$ are dominated by $O(M^2)$. Therefore the time complexity of this phase is $O(M^2)$.

In the probabilistic knowledge transfer phase, ants leverage pheromone information from related tasks. Each decision requires the selection of the source of the pheromone matrix, introducing a per-ant cost of $O(|V|)$. Across all ants and tasks, the total complexity for knowledge transfer is $O(NM|V|)$.

Summing up all these computational costs, the total time complexity per iteration of MTACO is $O(NM|V|^2 + M|E| + 2NM|V| + M^2) = O(NM|V|^2 + M^2)$. The final complexity expression equals to $O(GNM|V|^2 + GM^2)$.

Compared to conventional ACO, which has a time complexity of $O(GN|V|^2)$, MTACO introduces additional computational overhead due to multi-task knowledge transfer mechanisms and task similarity evaluation. However, these additional computations contribute to improved convergence speed and solution quality, making MTACO more efficient for solving large-scale MVPP instances. While MTACO has a higher per-iteration complexity, its ability to exploit task similarities reduces the total number of iterations required to reach an optimal solution, leading to overall efficiency gains in practical applications.

5 Experiments

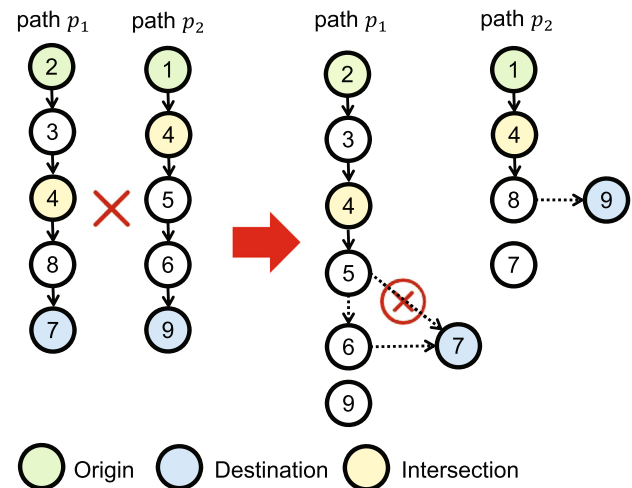
In this section, a comprehensive empirical study of MTACO is conducted by comparing its performance with three algorithms.

5.1 Comparison Designs

The performance of MTACO is compared with ACS [38], MTACO-L [39], and MTACO-DMSI [36]. ACS is a traditional single task ACO. It is mainly used to verify the effectiveness of the probabilistic knowledge transfer strategy.

Table 2 Parameters of the ACO

Parameter	Value
Number of ants (m)	6
Initial Pheromone Level (τ_0)	0.001
Pheromone Evaporation Rate (ρ and α)	0.7
Heuristic Function Weight (β)	0.7
Transfer Decision Factor (q_r)	0.3
Pheromone Matrix Decision Factor (q_s)	0.3
State Transfer Decision Factors (q_0)	0.5
Screening Radius (r)	100
Initial pheromone concentration weights (ε)	0.01

**Fig. 5** The knowledge transfer strategy for MVPP problems in DMSI

MTACO-L is equivalent to MTACO without archived data. It is mainly used to check the effectiveness of the archive-based initialization strategy. Archived data used for the initialization of MTACO is generated by executing test instances once.

Since the knowledge transfer strategy in MTACO-DMSI is not directly applicable to solving the MVPP problem, we adapt the knowledge transfer strategy to make it suitable to the MVPP problem and follow the MTACO-DMSI framework. The MTACO-DMSI adopts an explicit knowledge transfer strategy, where the elite solutions intersect at

the same vertices and subsequently detect the vertices that can reach the destination in the subsequent vertices, the strategy is explained in Fig. 5. In Fig. 5, two paths with different origins and destinations intersect at vertex 4. Path 1 maintains its original route up to vertex 4. After reaching vertex 4, it follows the route of Path 2, and it is evaluated whether the current vertex can lead to the destination vertex 7. For example, vertex 5 cannot reach the destination (vertex 7), while vertex 6 can reach vertex 7. The same process is applied to Path 2. As a result, the final paths are determined to be $\langle 2, 3, 4, 5, 6, 7 \rangle$ and $\langle 1, 4, 8, 9 \rangle$. MTACO-DMSI does not require a similarity measurement strategy because SNP tasks with different dimensions are highly similar. Therefore, ant colonies with the most similar tasks are used for knowledge transfer in MTACO-DMSI. This is the same as the idea in [36].

The parameters of the MTACO are listed in Table 2, where the parameter values for τ_0 , ρ , α , β , and q_0 follow [38]. Subsequently, the value of ε is set to be 0.01 so that it maintains a certain magnitude of adjustment over the initial pheromone concentration to strike a balance between exploration and convergence. In the experimental tasks of this study, there may be a high similarity between the tasks. If the value of r is too small, it may cause too narrow the range of knowledge sharing and affect the optimization performance. To cover all tasks as much as possible while ensuring effective knowledge transfer between tasks, r is set to be 100 to ensure full use of information from similar tasks and improve the overall optimization efficiency. In practical applications, it is rare to encounter multi-task scenarios with extremely high similarity. Therefore, we set the q_r and q_s to be 0.3 to ensure strong generalizability. This configuration balances inter-task information sharing while maintaining the independence of individual tasks, thereby achieving robust optimization performance across different levels of task similarity.

5.2 Test Instances

The transportation networks used in the test examples were categorized into two types, i.e., small-scale and large-scale. The small-scale network is called G_z -50, which has 50 vertices on the partial map of Guangzhou as shown in Fig. 6 (a). The large-scale network is called M -200, with 200 manually generated vertices as shown in Fig. 6 (b). The G_z -50 and M -200 network instances are utilized to compare the performance of the algorithms across different network scales. We generate three test instances for each network, each instance with 10, 30, and 50 tasks, respectively. Tasks in the test instance are executed at the same time. Table 1 lists the details of the test instances, including the number of tasks, the mean value and the standard deviation of the similarity among tasks, the average number of similar tasks, the transportation network, the number of vertices and links. Test

instances T_1 , T_2 , and T_3 called simple cases, test instances T_4 , T_5 , and T_6 called complex cases.

6 Comparisons and Analyses

6.1 Comparison of Results

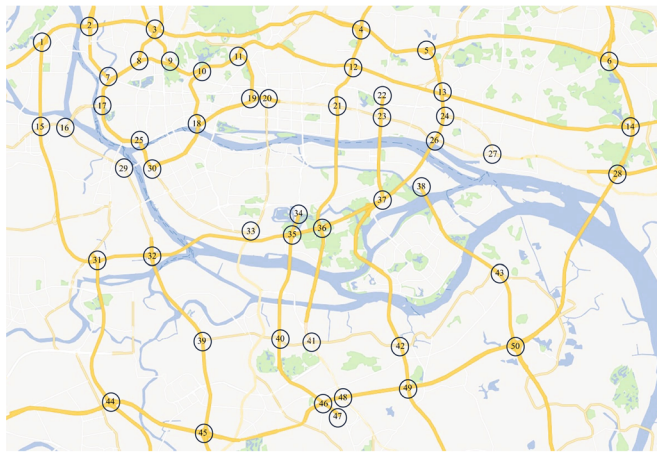
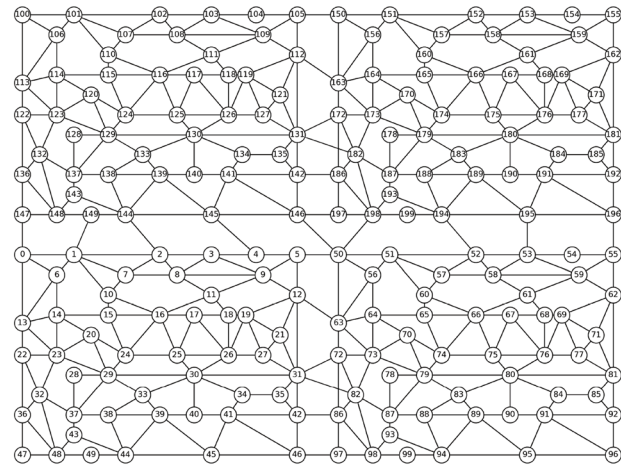
In this section, we compare the results of 20 independent runs for all algorithms. Table 3 lists the results of all compared algorithms, including both mean and standard deviation values, with the best results marked in bold. The $+$, $-$, and $\approx$ indicate that the algorithms are better than, worse than, or similar to MTACO respectively in the Wilcoxon signed-rank test of $\alpha = 0.05$.

All mean values of MTACO are better than or at least similar to other algorithms in all the test instances. In the simple cases T_1 , T_2 , and T_3 , all algorithms find an approximate optimal solution stably. Due to the low complexity of the tasks in the transportation network, all algorithms do not show a significant difference in terms of mean values. However, the knowledge transfer strategy still plays a role in speeding up the convergence of the algorithms in simple cases, which will be analyzed in section 6.2. In complex cases, the results of ACS are less satisfying compared to the other algorithms. This shows that the traditional ACO underperforms on complex MVPP problems. The mean values of MTACO-L are better than those of MTACO-DMSI, which shows that the probabilistic knowledge transfer strategy is effective and has greater advantages in the MVPP problem. The results of MTACO are better than MTACO-L. It indicates that endowing ACO with a priori knowledge is helpful for subsequent exploration. Based on the above results it can be concluded that pheromone-based implicit knowledge transfer improves ACOs in solving complex MVPP problems.

6.2 Comparison of Convergence

In this section, we compare the convergence curves of the four algorithms. Fig. 7 shows the convergence curves of the compared algorithms. In order to clearly observe the gap between curves, the x-axis of Fig. 7 (a)-(c) is from the first to the 200th generation because algorithms converge relatively quickly in simple cases.

Firstly, the positive impact of the probabilistic knowledge transfer strategy is demonstrated in the convergence trends of MTACO-L, MTACO-DMSI, and ACS in Fig. 7. We find that MTACO-L outperforms MTACO-DMSI overall in all simple cases, and outperforms MTACO-DMSI in complex cases only at later iterations. The probabilistic knowledge transfer strategy allows ant colonies with similar tasks to guide each other to explore new and promising paths, so it

(a) *Gz-50*(b) *M-200***Fig. 6** Test transportation networks**Table 3** The performance of the algorithms on 20 independently run test instances (the best results highlighted in bold. The +, −, and ≈ indicate that the algorithms are better than, worse than, or similar to MTACO in the Wilcoxon signed-rank test of $\alpha = 0.05$)

Instances		MTACO	MTACO-L	MTACO-DMSI	ACS
T_1	mean	269.63	269.63(≈)	269.63(≈)	269.6(≈)
	std.	0.00	0.00	0.00	0.00
T_2	mean	269.31	269.31(≈)	269.31(≈)	269.31(≈)
	std.	5.68e-4	5.68e-4	5.68e-4	5.68e-4
T_3	mean	263.99	263.99(≈)	264.00(≈)	263.99(≈)
	std.	0.00	0.00	0.03	0.00
T_4	mean	427.17	440.21(−)	444.73(−)	537.51(−)
	std.	4.71	8.74	7.15	6.52
T_5	mean	419.75	427.10(−)	436.29(−)	515.54(−)
	std.	3.05	3.32	3.63	5.22
T_6	mean	398.95	404.43(−)	415.53(−)	496.85(−)
	std.	0.93	2.91	3.11	3.50

relies greatly on the pheromone concentration of the source ant colony. In simple cases, the task complexity is low, so the ant colony obtains a lot of effective knowledge through just a few iterations. However, in complex cases, it is difficult for the ant colony to obtain enough useful knowledge through a few iterations, resulting in MTACO-L being weaker than MTACO-DMSI at the beginning. As the pheromone concentration increases, the knowledge shared among the ants becomes more reliable and useful, which makes MTACO-L gradually better than MTACO-DMSI.

Secondly, the final results and convergence speed of MTACO are all better than other algorithms, especially in large-scale transportation networks. The archive-based initialization strategy gives MTACO high-quality priori knowledge in the first generation, guiding ants to explore a promising search space. In contrast, the ACS, MTACO-L, and MTACO-DMSI lack priori knowledge, thus they can only explore blindly throughout the transportation network.

Finally, the archive-based initialization strategy and the probabilistic knowledge transfer strategy complement each other. The knowledge provided by the former in the initialization is used for knowledge sharing by the latter, which makes MTACO outperform MTACO-L and MTACO-DMSI in the selected test instances. Therefore, the implicit knowledge transfer strategy based on the pheromone matrix is showed to be effective.

6.3 Comparison of First-generation Fitness

In this section, we compare the fitness of MTACO, MTACO-L, and MTACO-DMSI at the first generation. Table 4 lists the standard deviation, mean, and the Wilcoxon signed-rank test of the algorithms for the first-generation fitness. Fig. 8 shows the distribution of their first-generation fitness over the six test instances.

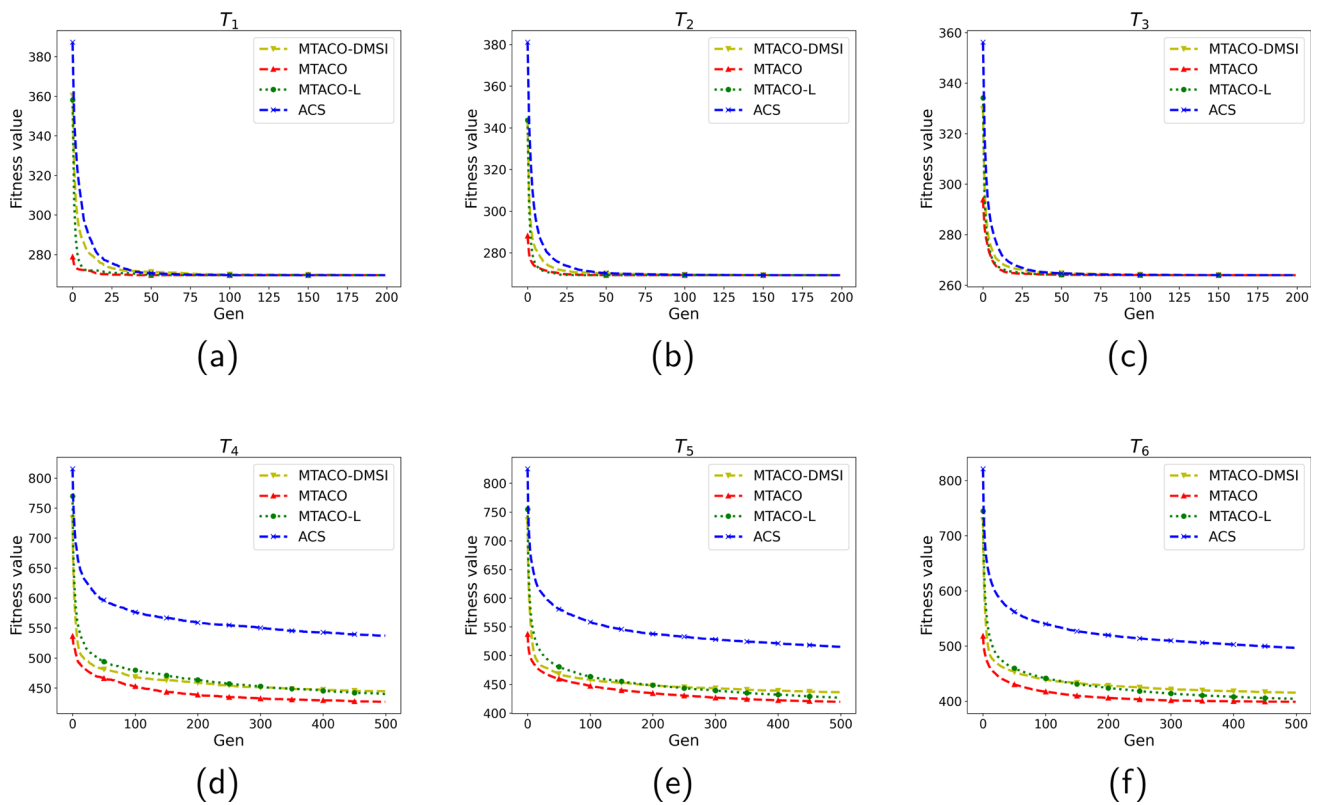


Fig. 7 Convergence curves for algorithms

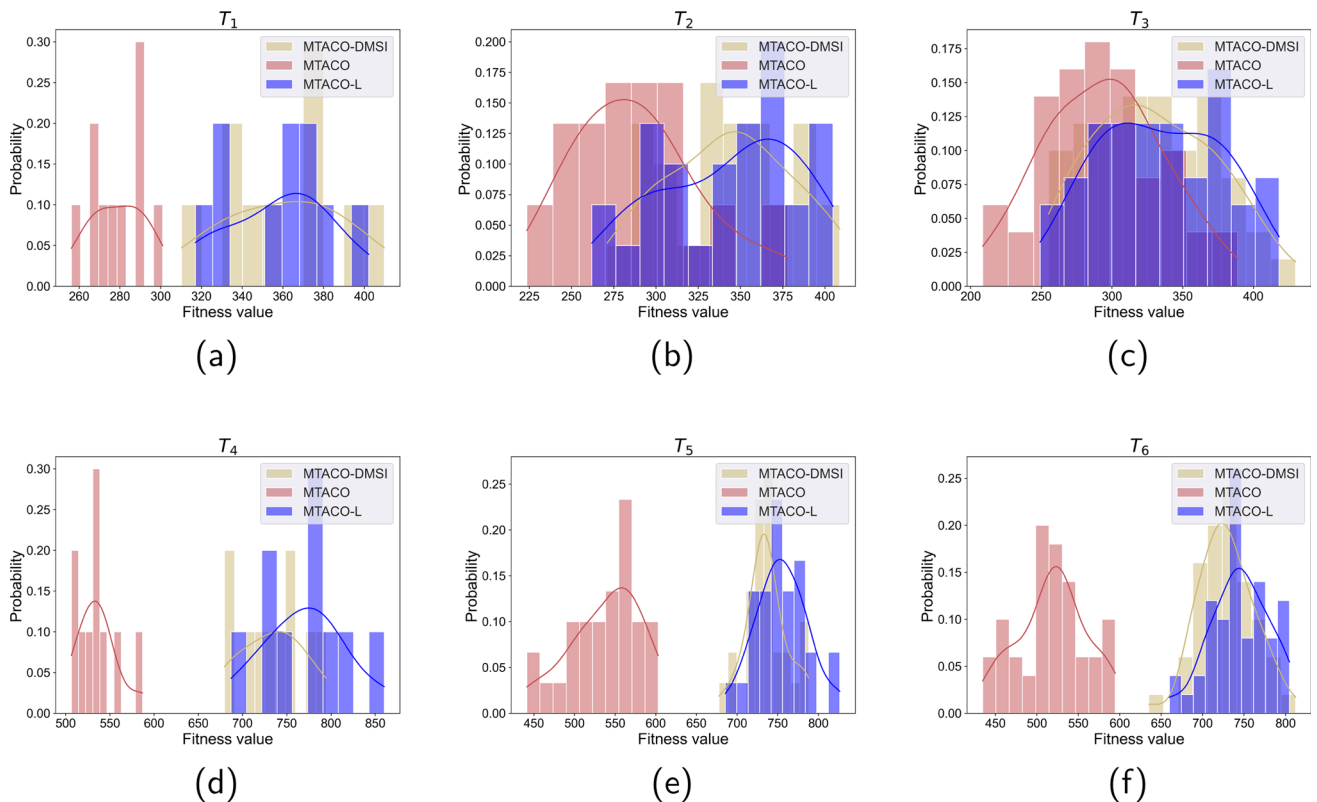


Fig. 8 Distribution of first-generation fitness for MTACO-DMSI, MTACO-L, and MTACO

Table 4 The first-generation fitness values of the algorithms on 20 independently run test instances (the best results highlighted in bold). The +, −, and $\approx$ indicate that the algorithms are better than, worse than, or similar to MTACO in the Wilcoxon signed-rank test of $\alpha = 0.05$

Instances		MTACO	MTACO-L	MTACO-DMSI
T_1	mean	278.92	358.13(−)	360.39(−)
	std.	4.31	26.58	18.35
T_2	mean	288.21	343.65(−)	343.09(−)
	std.	3.63	12.04	10.84
T_3	mean	293.84	334.19(−)	329.79(−)
	std.	3.82	11.53	7.30
T_4	mean	536.63	769.93(−)	733.33(−)
	std.	18.84	35.86	36.38
T_5	mean	537.61	754.92(−)	735.87(−)
	std.	12.99	26.99	23.22
T_6	mean	518.29	744.70(−)	728.59(−)
	std.	11.43	21.99	9.09

In terms of mean values, the quality of the first-generation solutions of MTACO is better than the other algorithms. Similarly, in Fig. 8, most of the first-generation fitness of MTACO are distributed in a better range (i.e., smaller values) than MTACO-L and MTACO-DMSI. The main reason is that MTACO utilizes the knowledge of archived data to guide guides the ant colony in a promising direction of exploration during the initialization phase. In other words, MTACO has gained relevant experience in the first generation, while MTACO-L and MTACO-DMSI are still aimless.

We observe that MTACO-DMSI slightly outperforms MTACO-L in terms of the quality of the first-generation solutions. The knowledge transfer of MTACO-DMSI is an explicit knowledge transfer based on elite solutions that is performed after all ant colonies have explored the path. Therefore, it is likely to obtain better solutions (lower fitness) in the first generation. Overall, the archive-based initialization strategy has a positive impact for the initial ant colony.

6.4 Parameter Analysis

The knowledge transfer parameters q_r controls whether an ant utilizes the pheromone matrix of its own task or that of a similar task. A lower q_r value prioritizes knowledge transfer from similar tasks, while a higher value encourages the ant to rely on its own pheromone information. When borrowing pheromone data from other tasks, q_s determines the selection mechanism. A higher q_s value enforces a strong preference for the most similar task, whereas a lower value allows the ant to randomly select from a set of similar tasks.

To analyze the impact of q_r and q_s , we conducted comparative experiments using values $\{0.0, 0.3, 0.5, 0.7\}$, with the MTACO algorithm solving the test instance T_4 on 16 param-

Table 5 Results obtained by MTACO in solving test instance T_4 with 16 parameter combinations

Parameters		$q_r = 0.0$	$q_r = 0.3$	$q_r = 0.5$	$q_r = 0.7$
$q_s = 0.0$	mean	442.35	458.56	464.22	458.18
	std.	18.16	21.77	20.56	21.20
$q_s = 0.3$	mean	442.50	448.86	452.13	457.33
	std.	17.44	18.89	19.60	23.19
$q_s = 0.5$	mean	442.81	455.27	463.69	464.45
	std.	18.25	19.58	20.27	18.20
$q_s = 0.7$	mean	457.60	454.18	463.29	462.26
	std.	20.85	19.01	19.93	21.77

eter combinations. The evaluation focused on average total path length and convergence speed. Table 5 lists the results, showing that mean values exhibit fluctuating upward trends as q_r and q_s increase.

Fig. 9 shows the convergence curves for different values of q_r while q_s is fixed, which demonstrates that higher q_r values consistently degrade solution quality by making ants over-rely on pheromone matrices of their own tasks, reducing diversity and increasing susceptibility to local optima. Notably, $q_r = 0.0$ represents an extreme case where ants rely entirely on pheromone matrices from similar tasks. Since the tasks in test instance T_4 exhibit high similarity, $q_r = 0.0$ performs overall well as the shared pheromone matrix provides effective guidance. However, when task similarity is low, knowledge transferred from other colonies becomes less useful. In such scenarios, the absence of task-specific pheromone guidance forces the target colony to explore aimlessly, reducing solution quality. Therefore, the value of q_r should be set according to the task similarity. For highly similar tasks, a lower q_r even near 0.0 is advantageous, enabling the colony to leverage effective shared pheromone matrices and discover superior solutions. Conversely, for low-similarity tasks, ants should prioritize their own task-specific pheromone matrix. Thus, q_r should set a value within $[0.3, 0.7]$ to balance exploration and self-guided optimization, ensuring adaptability to both shared and unique task characteristics.

Fig. 10 presents the convergence curves for different values of q_s while q_r is fixed, which reveals that $q_s = 0.3$ yields optimal stability by balancing focused knowledge acquisition from the most similar task with exploratory random selection. Higher q_s values enhance exploration but risk incorporating low-similarity task data, while lower values excessively restrict solution diversity.

Our analysis identifies $(q_r = 0.3, q_s = 0.3)$ as the most versatile configuration, optimally balancing exploration and exploitation across diverse scenarios. This combination maintains solution quality while adapting effectively to varying degrees of inter-task similarity.

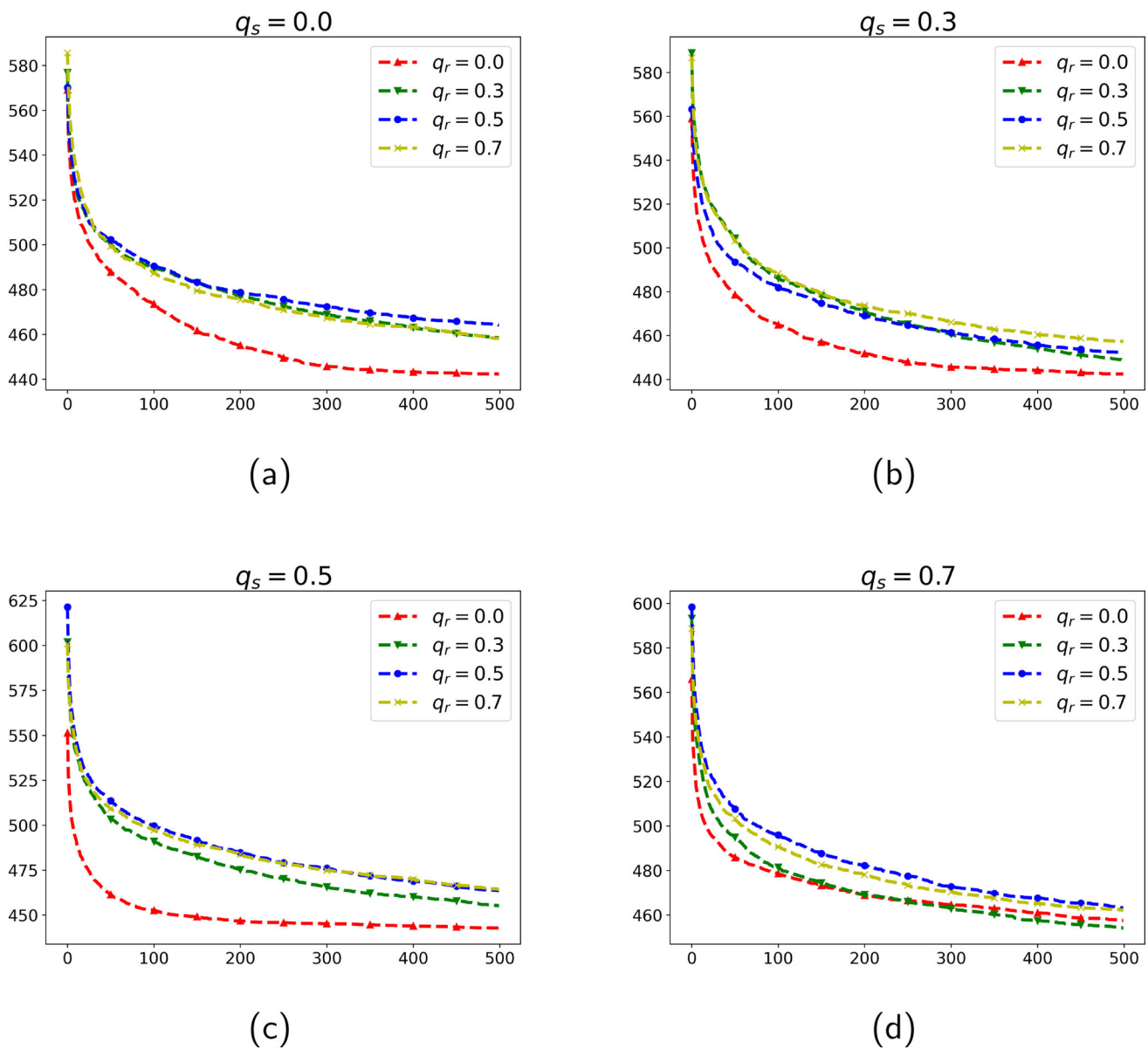


Fig. 9 Convergence curves for different values of q_s when q_r is fixed

7 Conclusion

This paper proposes a novel Multi-Task Ant Colony Optimization (MTACO) algorithm to solve the Multi-Vehicle Path Planning (MVPP) problem. Traditional Ant Colony Optimization (ACO) algorithms suffer from inefficient early-stage searching and limited knowledge transfer across tasks in multi-task scenarios. To address these challenges, MTACO introduces an archive-based initialization strategy, a probabilistic knowledge transfer strategy, and a similar-task association strategy. Specifically, the archive-based initialization strategy as a foundation for knowledge inheritance, ensuring that previous optimization results contribute to the

initial search direction. Meanwhile, the probabilistic knowledge transfer strategy enables parallel knowledge exchange, allowing multiple tasks to evolve collaboratively while maintaining search diversity. The similar-task association strategy further enhances knowledge relevance by ensuring that transfer occurs only between tasks with high structural similarity, thereby improving solution quality and convergence speed.

To validate the effectiveness of MTACO, we conducted experiments on six different MVPP test instances, comparing its performance with traditional ACO and other state-of-the-art algorithms. The results demonstrate that MTACO consistently achieves superior performance, particularly in complex problem settings, highlighting the advantages of

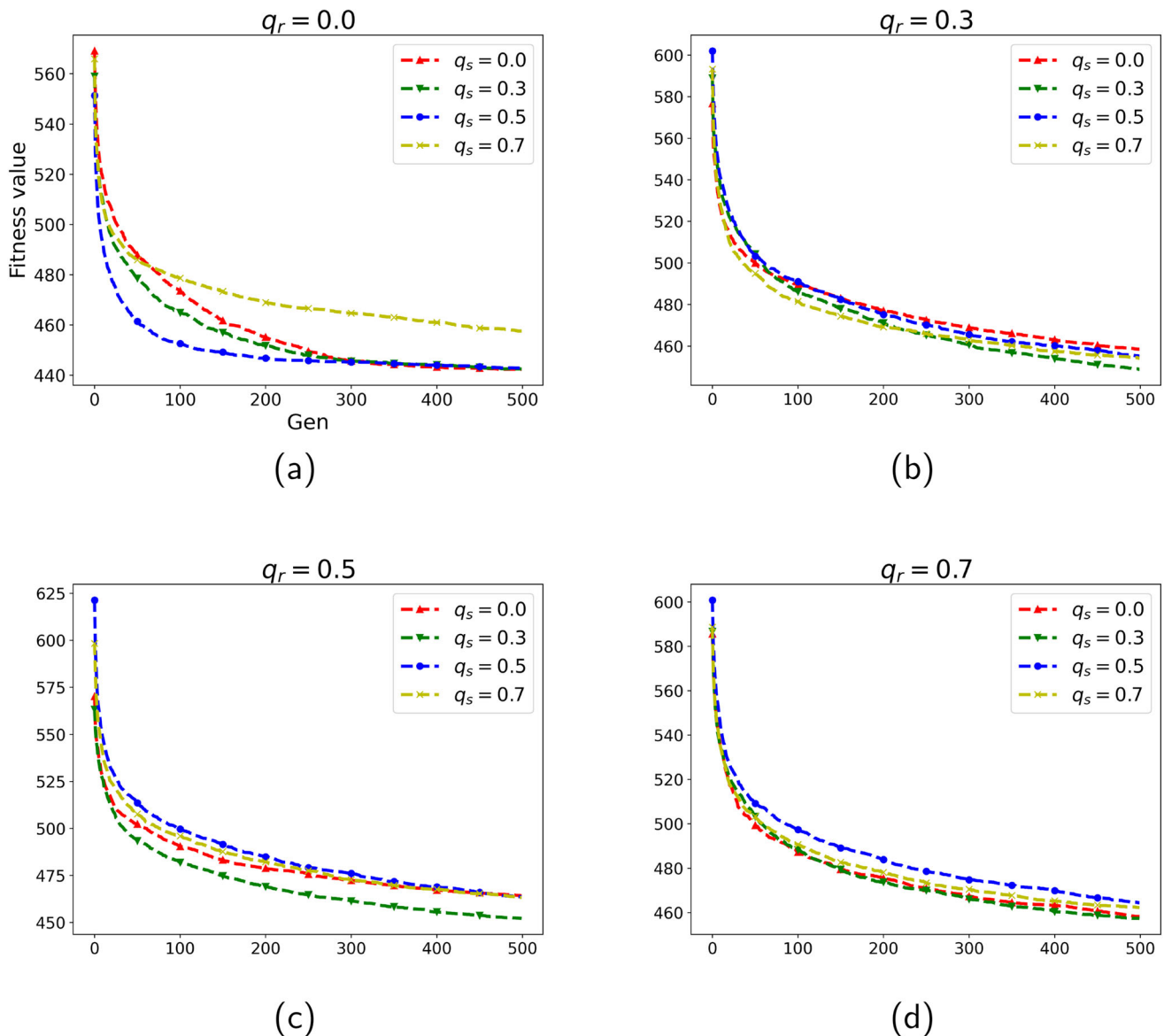


Fig. 10 Convergence curves for different values of q_r when q_s is fixed

integrating multi-task optimization principles into ACO. By leveraging historical knowledge, inter-task pheromone sharing, and similarity-based knowledge transfer, MTACO significantly enhances optimization efficiency. However, the performance of our proposed MTACO is affected by the quality of archived solutions, task similarity, parameter sensitivity, and computational overhead. Future research should focus on adaptive learning mechanisms, improved similarity evaluation, and enhanced dynamic adaptation to further strengthen MTACO's capabilities. With continued refinements, MTACO has the potential to contribute to a wide range of combinatorial optimization problems.

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Declarations

Competing Interests All authors certify that they have no affiliations with or involvement in any organization or entity with any financial

interest or non-financial interest in the subject matter or materials discussed in this manuscript.

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